Performance Improvements Through Advanced PV Backtracking on Uneven Terrain

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***Abstract*—The climatic sensitivity of new terrain-aware backtracking algorithms is evaluated across 800 locations in the continental USA on a representative synthetic rolling terrain. We find that a global optimization approach to backtracking results in climate-specific annual energy gains of 2.4%–3.2% relative to a traditional backtracking algorithm baseline. We identify a strong logarithmic correlation between local diffuse fraction and yield improvement, and highlight the effect of seasonal precipitation on performance gains. We also find that a backtracking approach, which approximates the terrain as constant, does not offer significant annual energy gains over the baseline on the synthetic terrain. Our findings suggest that specific yield from backtracking in the USA can be improved by as much as 88 kWh/kW by considering terrain when selecting a backtracking algorithm.**

***Index Terms*—Backtracking, performance modeling, shading, single-axis tracking.**

I. Introduction

In the last eight years, tracking technologies have become ubiquitous in new ground-mounted photovoltaic (PV) installations. Of the 32.3 GW of utility-scale solar (defined as >5 MW ac) that came online in the USA between 2020 and 2022, 91% of capacity was mounted on single-axis tracking (SAT) systems [1]. SAT increases solar insolation collected by rows of PV panels by controlling the angle at which the panels are oriented relative to the Sun's position. SAT systems have torque tubes running N–S, each rotating in place to follow the position of the Sun from E to W, such that the angle of incidence between the PV and the Sun's rays is minimized (known as "true-tracking"). Self-shading is typically incurred by true-tracking algorithms near sunrise or sunset, when panels move to a more vertical orientation to face the Sun that is low in the sky.

Backtracking algorithms adjust the axis rotation angle away from true-tracking to avoid self-shading by panels moving to a more, horizontal (i.e., flat) orientation. Traditional backtracking algorithms assume graded, flat terrain. However, there is a common conception within the PV industry that utility-scale deployment is limited by the availability of flat land, which has prompted a perceived need for tracking technologies that can be deployed on uneven ground.

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Tracker manufacturers have risen to meet this challenge. Within the last few years, many manufacturers have introduced new "terrain-following" tracker products which claim to optimize PV production on ungraded or minimally graded terrain. Such terrain-following products include articulating joints and new backtracking algorithms that purportedly minimize row-to-row shading on uneven terrain. Unfortunately, the proprietary nature of these algorithms prevents them from being independently evaluated. Evaluating open-source offerings can provide insight into how general classes of algorithms perform.

Across open-source backtracking algorithms, there are significant differences in how uneven terrain is accounted for to determine a tracker's optimal rotation. Several algorithms use tracker geometry and solar position to find an analytic solution on the sloped terrain [2], [3], but assume that the terrain's cross-axial tilt is constant, as is defined in [3]. For a tracker with rows running N–S, cross-axial tilt would be the slope of the terrain in the E–W direction.

The simplicity of analytic solutions is appealing, leading to their adoption in some commercial trackers as well as the PV system performance modeling software [4], [5]. However, their assumption of constant terrain slope limits their real-world applicability to naturally planar terrain or terrain subjected to grading. In practice, the rarity of naturally planar terrain and the cost associated with grading motivate more sophisticated backtracking methods that can accommodate variations in terrain slope [6].

Two open-source backtracking methods designed for rolling terrain (i.e., terrain with varying slope) have been recently published; one is a closed form solution [7], and the other formulates backtracking as a linear programming (LP) optimization problem and determines an optimal backtracking strategy for the entire array in a single iteration [8]. The latter optimization seeks to point each tracker row to collect the maximum solar irradiance while preventing shading on subsequent rows. Thus, the entire system of trackers captures the most energy possible while avoiding the high energy penalties associated with shading PV modules. Using freely available open-source LP solvers [9], this method claims to be efficient enough to determine backtracking rotations for thousands of tracker rows in less than a second of computation time.

There have been several studies that quantify the performance of backtracking algorithms on terrain with a constant slope [10], [11], but studies of performance on rolling terrain have thus far been limited to commercial backtracking methods [6], [12]. In this study, we seek to fill this gap by comparing the performance of several open-source algorithms on rolling terrain.

In this study, we quantify performance gains from two analytic approaches [3] and one LP method [8] relative to a baseline algorithm [13]. We control for the impact of terrain on performance gains by introducing a synthetic rolling terrain based on sites of recently deployed utility-scale systems. We quantify the impact of seasonal weather patterns on performance gains by simulating tracker performance at over 800 locations across the USA, and identify regions that experience the highest and lowest performance gains. We connect performance gains to climate characteristics and isolate the effect of seasonality on gains.

II. Methods

A. Terrain and System

Terrain variation is, by definition, extremely local. In the context of PV plants, which might occupy a few square kilometers, the terrain may change greatly even a few tens of kilometers away from a given point. In order to isolate the effects of tracking algorithm and site weather patterns, while using a consistent and repeatable terrain, we developed a synthetic terrain profile to be used at each of the sites. The consistent use of a synthetic terrain allowed for comparisons of the tracking algorithms and weather while eliminating differences caused by variations due to highly localized terrain at a site. While this terrain is not identical to any particular terrain, it is designed to be a representative of a terrain that is commonly used to develop large PV systems as described below. The resulting terrain is used here as a control element to investigate the variables of tracking algorithm and site weather. We expect that the terrain with larger elevation variation would, in general, show an increase in performance gains for optimized backtracking algorithms.

The synthetic terrain is flat in the N–S direction and varies in the E–W direction according to (1). The effect is a mostly-W-facing slope with a maximum 18 m of elevation change within the 2000 m span. There are also small sinusoidal variations with periods of 157 m

$$y = c_1 x + c_2 \cos(c_3 + c_4 x) + c_5 \sin(c_6 + c_7 x) \quad (1)$$

where $c_1 = 0.00001, c_2 = 4, c_3 = 0.1, c_4 = 0.004, c_5 = 0.08, c_6 = -0.1$, and $c_7 = 0.04$.

This terrain was designed to be similar to the terrain on which utility-scale systems are currently being deployed. Using the locations of utility-scale systems listed in the USA Large-Scale PV Database (USPVDB) published by Lawrence Berkeley National Laboratory and the USA Geological Survey (USGS) [14], we retrieved 2000 m $\times$ 2000 m digital elevation maps (DEMs) published by the USGS with 1 m resolution. As 1 m DEMs are not available for the entire USA, this restricted our analysis to 428 sites of the 903 total sites listed in the USPVDB. E–W slopes for these locations are ranged from 0° to 10°, with most of the systems having an average slope between 0° and 4°. The synthetic terrain used in this study has a slope which ranges between $-0.5°$ and $1.5°$; the variance in the slope of the synthetic terrain is also consistent with that of the locations associated with systems in the USPVDB.

A representative SAT with torque tubes oriented in the N–S direction was modeled along the E–W axis of this terrain at a

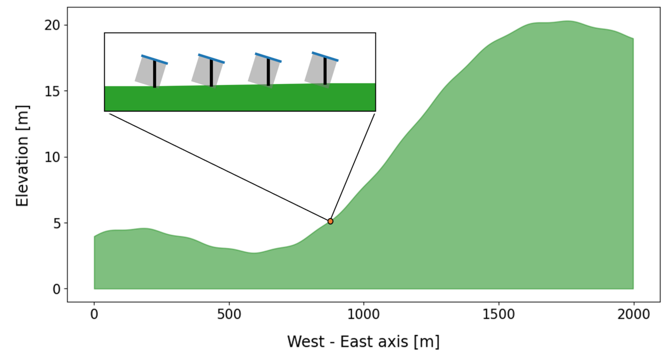

Fig. 1. Synthetic terrain used in simulations. Note the different axes scales.

4.5 m pitch for a total of 445 rows. Monofacial PV modules with full-cut cells and a conventional three-string interconnection design are oriented in portrait mode and stacked one panel high along each tracked row such that the E–W edge of each row is 2 m long. The rows' rotations are restricted to a maximum tilt of $\pm 50°$ from horizontal. Each row is free to rotate independently of other rows.

B. Backtracking Algorithms

Four different backtracking algorithms were simulated for the synthetic terrain. Two were implementations of the "slope-aware" method published in [3] and implemented in pvlib-python [5] while one was an implementation of an LP approach [8]. Performance was benchmarked against a conventional backtracking approach that assumes a flat, horizontal terrain [13]. This algorithm assumes that the goal is to avoid row-to-row shading entirely, which is suitable for modules of full-cut cells as assumed here, but may not be the optimal strategy for modules of half-cut cells [7], [15].

The "slope-aware" method [3] is similar to conventional backtracking in that it first finds the true-tracking angle using tracker geometry and solar position, and then makes an adjustment to account for self-shading. Our first implementation of this method used the average slope across the entire synthetic terrain ($\approx 0.6°$) to determine the rotation angle for all rows of the modeled system and is referred to as an *analytic solution with global parameters* (see Fig. 2). In our second implementation, rotation angles were calculated separately for each row based on the slope of the terrain directly beneath the row; this is, the *analytic solution with local parameters*.

The LP approach formulates the problem by maximizing the sum of direct irradiance received by all rows [8]. The formulation is implemented by enforcing an inequality on the direct irradiance received by adjacent rows and the total available direct irradiance. This approach will be referred to as the *global optimization method*.

The baseline algorithm is the most likely to cause row-to-row shading or leave uncollected irradiance between the rows over sloped ground. The analytic solution with global parameters is likely to perform well when the slope under a tracker row is similar to the overall slope of the site. The analytic solution with local parameters is less likely to cause row-to-row shading or

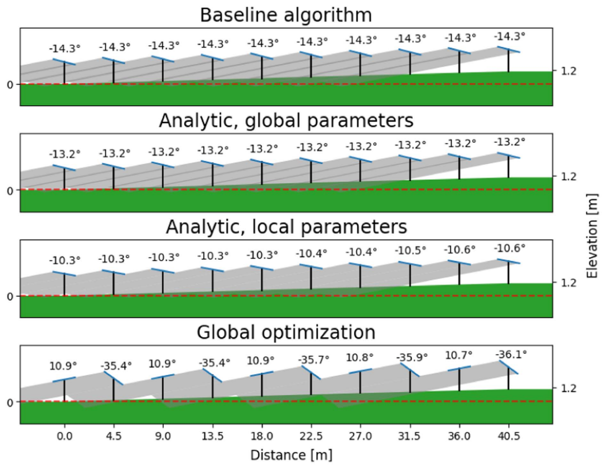

Fig. 2. Tracker rotations modeled using algorithms on a cross-section of the synthetic terrain spanning approximately 40 m.

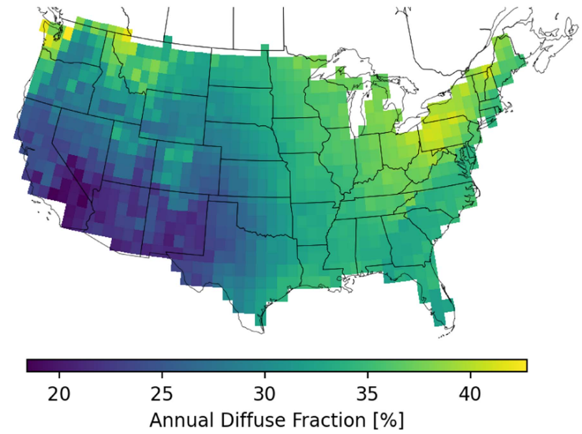

Fig. 3. Annual diffuse fraction is defined as the ratio between DHI and GHI integrated over a year.

leave uncollected irradiance, while keeping the trackers tilted in a similar direction. The global optimization method is designed to eliminate row-to-row shading and maximize collected irradiance, but may choose tracker angles that, while consistent over time, may not have trackers pointing in generally the same direction as shown in Fig. 2.

Tracker angles were calculated based on the projected solar zenith angle (PSZA), which is the zenith angle in the tracker's frame of reference, a metric introduced in [3]. PSZA is calculated using solar position and tracker geometry. Although there are seasonal variations in its rate of change, PSZA sweeps from -90° to 90° each day regardless of location or time of year.

The differences in tracking between the baseline algorithm and the alternative algorithms are observed only when the PSZA is near -90° or 90°. It is at these times that variations in terrain may cause backtracking rows with the baseline algorithm to cast shadows on rows behind them. Gains or losses when the PSZA is near 0° are unlikely, as all of the algorithms perform similarly.

C. Performance Modeling

Five years of 30-min resolution weather data (2018–2022), including direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), global horizontal irradiance (GHI), air temperature, wind speed, and surface air pressure, were retrieved from the national solar radiation database [16] for over 800 locations in the continental USA at $1^\circ \times 1^\circ$ resolution and interpolated to a 5-min resolution (see Fig. 3).

1) Near Horizon Shading: For each tracker row in the model, a unique horizon profile was generated to determine when the Sun would be obscured by the terrain. All terrain points to the east and west of each row were considered and the maximum PSZA which did not intersect the terrain was determined. When the sun PSZA exceeded the eastern or western terrain horizon, the given tracker row was considered to be shaded by the terrain and the DNI for the tracker row was set to 0 and the GHI was set to DHI, thus simulating the blocking of direct solar irradiance by the terrain. The maximum PSZA, which did not cause horizon shading, was limited to 90° to indicate that the far-horizon is still a limiting factor that always shades a given tracker row.

2) Power Modeling: Plane-of-array (POA) irradiance for each row was modeled using the Hay–Davies sky diffuse model [17]. Effective irradiance (Ee) was modeled using the sandia array performance model (SAPM) [18], which includes irradiance corrections for spectral mismatch and an incident angle modifier. Cell temperature (Tc) was also modeled using the SAPM using thermal coefficients for a 320 W 72 cell mono-Si module. Power was modeled for each row using the PV Watts model [19] with Ee and Tc. As power is proportional to Ee in the PVWatts model, power was then multiplied by a row-specific derate factor to account for decrease in direct irradiance due to interrow shading (see Section II-C3).

3) Interrow Shading: For each tracker row the interrow shading from adjacent trackers was also considered separately from near-horizon shading. This calculation evaluated the rotation position of the nearest five tracker rows in the direction of the Sun and determined the amount of shade each of those five adjacent rows cast on the tracker row being evaluated. The shaded fraction of the target tracker was calculated using the procedures given in [7]. The decision to include only the five nearest trackers, rather than the entire array of trackers, was made to reduce computational effort and based on the intuition that the nearest five trackers were the most likely to shade the evaluated row. When a tracker row was shaded by a neighboring tracker, it was assumed that the entirety of the tracker row was shaded (i.e., edge-effects were ignored, simulating long tracker rows). Then, derate factors were calculated to reduce the output of the tracker row according to the shaded fraction in a piecewise procedure outline in [10, eq. (4)]. To summarize briefly, a tracker with nonzero shaded fraction less than one row of cells in the portrait oriented module (i.e., $1/12$ for a 72-cell module), a linear power loss is assumed with 0% loss of DNI at 0 shaded fraction and 100% loss of DNI when the shaded fraction is $1/12$ of the module (i.e., the first row of cells is shaded); for shaded fractions larger than $1/12$, it is assumed that the row's output power is generated using only diffuse POA irradiance. While this is a simplified model for shading losses, it is much faster to compute

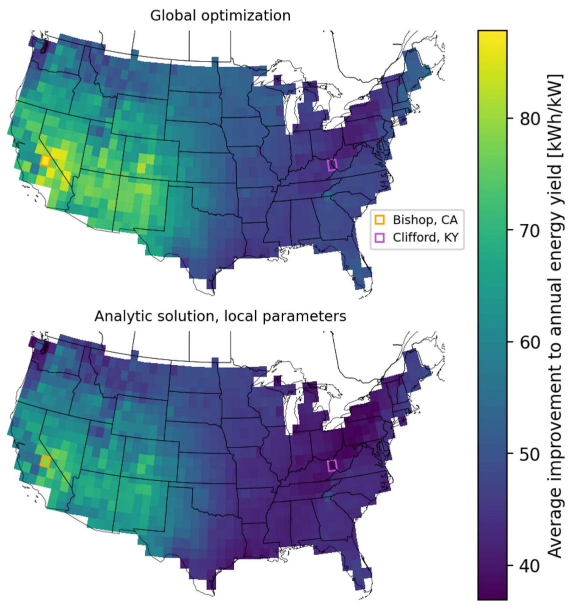

Fig. 4. Even regions that experience the lowest SY relative to the baseline still would benefit from an additional 40 kWh/kW increase in energy output from using either the global optimization method or the analytic approach with local parameters. Bishop, CA, and Clifford, KY, are labeled with colored boxes (see Section III-B for case study analyses).

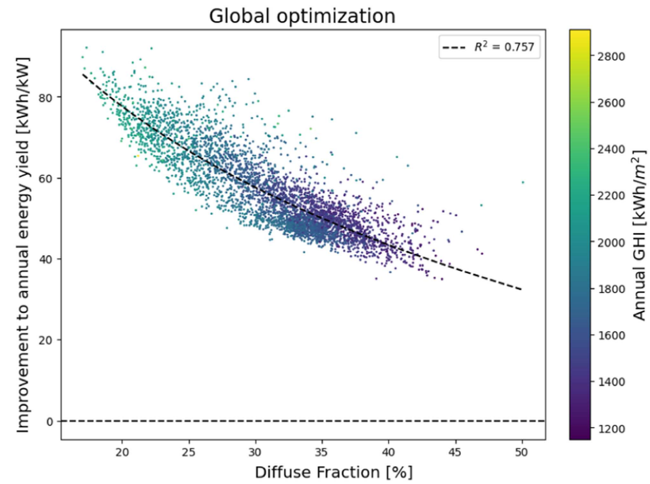

Fig. 5. Dependence of SY improvement on diffuse fraction is well-described by a logarithmic function.

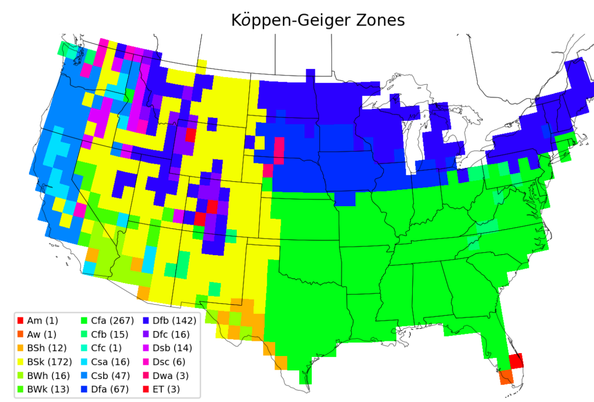

Fig. 6. Köppen–Geiger climate zone classifications at 100 arc-seconds. Zones are defined by patterns of seasonal precipitation and temperature. For example, zones with "f" do not experience a dry season, while zones with "w" or "s" have dry winters and dry summers, respectively. The number of $1^\circ \times 1^\circ$ locations in the USA, classified as each climate zone is noted in parentheses.

for many time series and locations used in this article than complex shading models. To estimate the effect of this simplified shading loss model, we modeled the proposed SAT system over the year 2020 in Kansas City, MO, using the simplified shading model and a more complex single diode model to estimate shade losses. The baseline tracking algorithm was used for this analysis since it is the most likely algorithm to cause shaded modules. The simple model of shading estimated 2.44% of shade losses over the year while the complex single diode model estimated 1.78% shading losses over the year. We conclude that the simple shading model tends to overestimate the effect of shading. Thus, the energy gains available via the advanced backtracking algorithms presented here may be slightly overestimated compared to the gains that may be available in fielded systems.

III. Results

A. National Trends

Averaged annual energy gains relative to the baseline over a five-year period were 2.8% with the global optimization approach and 2.5% with the analytic method with local parameters. The analytic method with global parameters generated negligible energy gains in most locations, and underperformed in some locations.

Specific yield (SY) gains relative to the baseline over a five-year period ranged from across the USA for both the global optimization method and the analytic approach with local parameters (see Fig. 4).

SY gains were inversely correlated with diffuse fraction (see Fig. 5), with small differences in diffuse fraction generally leading to significant improvements in SY. It is expected that the improvement of SY is highly dependent on terrain. However, further analysis of algorithm performance on other synthetic terrain segments is needed to quantify how diffuse fraction sensitivity scales with terrain.

Gains in SY (kWh/kW dc) were highest in the southwest and lowest in the midwest and southeast for both the global optimization method and analytic approach with local parameters (see Fig. 4).

The demarcations between regions with higher and lower SY gains are correlated with Köppen–Geiger climate zone classifications [20] (see Fig. 6). A Mann–Whitney test of gains found that they were greater in the west (BSk) than in the midwest and southwest (Dfa, Dfb, and Cfa) with a 95% confidence level (see Fig. 7).

SY gains on the west coast (Csb) were also greater than in the midwest and southwest. Both Csb (temperate-dry summer-warm-summer) and BSk (dry-steppe-cold) are marked by dry seasons while Cfa (temperate-no dry season-hot summer) and Dfa (continental-no dry season-hot summer) do not experience

TABLE I
IMPROVEMENTS IN SY WITH TRACKING ALGORITHMS FOR REPRESENTATIVE LOCATIONS, ORDERED BY BASELINE SY

Location	Annual Diffuse Fraction	Climate Zone	Annual baseline SY [kWh/kW]	Annual SY gain with global analytic solution [kWh/kW]	Annual SY gain with local analytic solution [kWh/kW]	Annual SY gain with global optimization [kWh/kW]
Portland, OR (45.5°N, 122.5°W)	37.9%	Csb	1548.4	-1.20	41.90	46.68
Albany, NY (42.5°N, 73.5°W)	36.6%	Dfb	1695.0	0.60	43.78	48.71
Clifford, KY (38.0°N, 82.5°W)	37.2%	Cfa	1723.9	-0.46	37.77	41.78
Kansas City, MO (39.0°N, 94.5°W)	33.4%	Cfa	1871.4	-1.20	44.57	50.03
Orlando, FL (28.5°N, 82.5°W)	33.1%	Cfa	2144.7	2.53	45.44	50.01
Albuquerque, NM (35.0°N, 106.5°W)	28.4%	BSk	2432.5	1.45	64.93	72.73
Bishop, CA (37.0°N, 118.5°W)	21.3%	Csb	2708.1	2.60	78.96	87.83

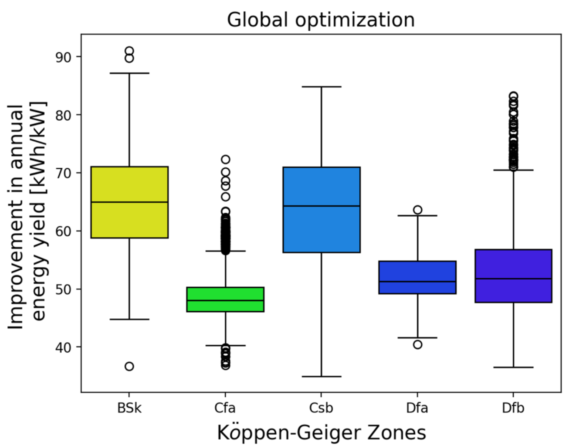

Fig. 7. SY improvement by zone.

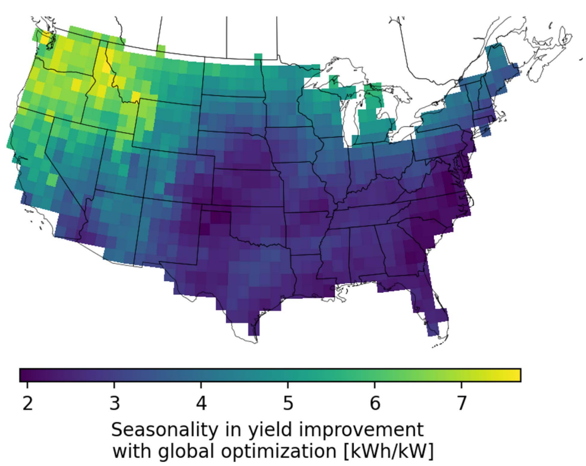

Fig. 8. Although seasonal variation in both yields improvement and GHI track closely with latitude, seasonal variation in SY improvement is especially pronounced in the Pacific northwest.

seasonal variations in precipitation. Given the different gains in drier (BSk and Csb) and wetter regions (Dfa, Dfb, and Cfa), it appears that SY gains are strongly correlated with precipitation. Precipitation necessitates cloud cover, which would be consistent with the strong correlation between the diffuse fraction and SY gain shown in Fig. 5.

An analysis of the seasonality of SY gain (largest monthly difference) confirms the influence of cloud cover. In general, seasonal differences in SY gains increase with latitude (see Fig. 8). Notably, Washington, Oregon, and Idaho (partially classified as Csb) see greater disparities between seasonal minimums and maximums in SY improvement than states that are further east (ex: Dfb). This difference is worth highlighting, as Csb and Dfb are similar in several major aspects. They are at roughly the same latitudes and both experience warm summers, although average monthly temperatures do not exceed 22 °C. The key difference between them—and what likely is driving the difference in their seasonal patterns of SY improvement—is their seasonal precipitation patterns; Csb has a dry season (summer) while Dfb does not. The lack of seasonal variations in cloud cover patterns, which is a characteristic of the Dfb region, likely generates consistently higher diffuse fractions, and consistently lower SY improvements.

B. Case Studies

Specific metrics for algorithm performance are reported for several representative locations in Table I and reflect macro trends described in Section III-A. Locational diffuse fractions listed in Table I are inversely correlated with SY for both the local analytic solution and the global optimization approach, consistent with Fig. 5. In contrast, significant variations in algorithm performance for locations within a single climate zone listed in Table I underscore the limitations of relying on zonal analyses to predict performance; of the locations listed in Table I, Portland, OR, and Bishop, CA, have the lowest and highest SY for each of the four algorithms, respectively.

IV. DISCUSSION AND CONCLUSION

A. Summary

Using a synthetic rolling terrain, gains in SY relative to the baseline were quantified for three different backtracking algorithms for over 800 locations across the continental USA. Two algorithms demonstrated substantial benefits in every location they were deployed, with annual SY improvements ranging from 40–90 kWh/kW, while the backtracking approach, which used the average slope for the synthetic rolling terrain, did not outperform the baseline. To put this into context, for a 100-MW PV plant with a power purchase agreement (PPA) price of $35/MWh [21], the potential added value of using these algorithms would be between $140 000 and $315 000 annually. We find that Köppen–Geiger climate zones correlate to SY improvement, and specifically highlight the influence of cloud cover on algorithm outcomes. Interactions between seasonal weather patterns and terrain variation remain largely unexplored, but could lead to extreme localized effects on tracker performance. For example, the effect of regular diurnal monsoonal patterns on performance may significantly differ depending on the timing of rainfall and whether the slope is east-facing or west-facing.

When averaged over all modeled USA locations, we find that optimally controlling SAT backtracking on sloped terrain can improve annual energy gains by 2.8%. If we look only at the Albuquerque, NM site over the five-year test period, we find that optimal tracking yielded 2.9% more energy than the baseline case. These findings align well with the results in [12] which modeled energy gains of 2.70% for Albuquerque based on typical meteorological year (TMY) data and correcting the model using the results of a field test with tracking that accounted for slope. However, we note that the location in [12] utilized a relatively flat terrain with a 1.1° slope to the east while our simulated terrain was more varied slopes between −0.5° and +1.5° to the east. The algorithm in [12] is not detailed, since it is a proprietary algorithm, but the agreement between the findings indicate optimization for terrain may yield a significant amount of additional energy on variable terrain.

The approach and results presented in this study may serve as the first step in a value assessment of terrain-following trackers in utility-scale systems. This study provides an approach to calculating the energy generation of an optimally tracked PV system, which is an important input to financial metrics (e.g., return on investment (ROI), levelized cost of electricity (LCOE), net present value (NPV), etc.).

B. *Limitations of This Work*

This work uses a simulated terrain with a gradual overall slope in the east-west direction. The results of this work are therefore specific to the simulated terrain, while investigating the effects of location and climate on this terrain. The effect of different tracking mechanisms and location may be different for different terrain. For example, if the terrains were to have higher slopes and more variations, optimal tracking may yield an even larger increase in SY compared to the baseline case.

This work models system performance using full-cell monofacial PV modules. While this module architecture represents the majority of currently-installed PV systems, most large PV systems being currently installed have bifacial and/or half-cut cells. In relative terms, the authors expect that using half-cut cells would decrease the simulated gain by approximately 50%, and bifaciality to decrease the simulated gain by approximately 10%, with the effect stacking for modules with both properties.

The authors also note that the simplified shading model used in this work overestimates energy losses (and thus potential energy gains) compared with a more detailed cell-level *I–V* curve simulation. In this study, the computational burden of the cell-level model was impractical, necessitating the use of the simpler model. However, where practical, future work could use the higher accuracy model instead.

C. *Future Work*

The authors have identified potential topics for future exploration to build upon this work and addressed the limitations noted above.

Future work should quantify the effect of terrain variability on performance over a range of terrain profiles. Specifically, such an analysis would benefit from testing on synthetic terrain segments. The performances of the four algorithms addressed in this study may diverge to greater or lesser degrees on different synthetic terrains; future research is needed to characterize algorithm performance on a larger range of terrain types. Furthermore, including terrain which varies *along* the tracker's rotation axis may more accurately simulate real-world gains expected from optimal backtracking technology.

Future work may also quantify the effect of different PV module architectures, such as bifacial and half-cut cells, and different PV system architectures, such as using trackers with groups of rows that must have identical tilt angles due to mechanical linking (or "ganging"). Comparing against backtracking algorithms tailored for modules of half-cut cells (e.g., Nextracker's "split-boost" algorithm [15]) is also of interest.

Finally, the authors find that the costs of terrain-following trackers—specifically the cost savings with decreased grading—are harder to quantify. Terrain-following trackers tend to be more expensive than traditional trackers, but manufacturers claim that the decreased grading required for these new products can justify their use. Grading costs are highly variable from site to site and are rarely shared publicly. The limited data published on this topic suggests that site preparation (which includes grading along with other earthmoving activities) costs less than 1¢ per W dc on average [22]. However, the authors' conversations with developers have suggested that the cost of site preparation can reach 9¢ per W dc. Future work toward an LCOE calculation should focus on obtaining data on grading costs, as well as specifics on how much grading is required for terrain-following products.

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